

Module/Pathway/Taxon Review: Bacterial Glycolate and Glyoxylate Assimilation in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Target bucket:** KEGG [ppu00630](#) — Glyoxylate and dicarboxylate metabolism
Module area: central_carbon **Scope reviewed:** glycolate → glyoxylate → glycerate assimilation module (phosphoglycolate dephosphorylation; multicomponent glycolate oxidation; glyoxylate carboligase / tartronate-semialdehyde branch)

1. Executive Summary

The core bacterial glycolate/glyoxylate-assimilation module is **genomically present and experimentally validated** in *P. putida* KT2440, but it is **conditionally active** because the key branch is transcriptionally repressed in the wild-type strain. Direct KT2440 evidence establishes that the GlcDEF glycolate dehydrogenase (PP_3745–PP_3747) oxidizes glycolate to glyoxylate, and that a co-transcribed operon — *gcl-hyi-glXR-ttuD-pykF* — carries glyoxylate to glycerate via glyoxylate carboligase Gcl (PP_4297) and tartronate-semialdehyde reductase GlxR (PP_4299). Overexpression of only *gcl* plus *glxR* is sufficient to confer growth on ethylene glycol, demonstrating that module steps 2, 3, and 5 are functionally covered. Consequently these steps should be marked **covered**, with the important caveat that the entire *gcl* branch is repressed by the regulator **GclR** in wild-type KT2440 and becomes active only upon regulatory de-repression (e.g., loss-of-function *gclR* mutations selected during adaptive laboratory evolution).

Two curation corrections are required. First, the generic "**step 1**" **2-phosphoglycolate phosphatase** (candidate PP_0416 / PP_1907) does **not** represent the physiological entry to glycolate assimilation in this organism. 2-Phosphoglycolate salvage is biochemically tied to the RuBisCO oxygenase reaction and the Calvin-Benson-Bassham (CBB) cycle / oxygenic photorespiration, which KT2440 — a chemoheterotroph — lacks. In the experimentally characterized KT2440 route, glycolate arises from oxidation of ethylene-glycol-derived

glycolaldehyde, not from 2-phosphoglycolate. This step should be flagged **module_needs_revision / not_expected_in_target_taxon** as the physiological entry, retaining PP_0416 only as a housekeeping/detoxification phosphatase.

Second, the **62-gene candidate list is dominated by neighboring-pathway "bucket noise."** KEGG map 00630 is a broad hub map that overlaps the glyoxylate shunt, TCA cycle, glycine/serine and C1 metabolism, nitrogen assimilation, and thiolase chemistry. Only ~9 candidates are true `ppu00630`-primary, and only ~7 genes form the actual glycolate→glyoxylate→glycerate module core (*glcDEF*, *gcl*, *hyi*, *glxR*, plus the PGP step). Reductase paralogs *ttuD*, *ghrB*, and *hprA* carry broad EC mappings and are likely over-annotated for this module; PP_0094 (5'-nucleotidase) is a clear map artifact with no role in glycolate/glyoxylate assimilation. No dedicated glycolate/glyoxylate permease is present among the candidates, consistent with an intracellular substrate-supply mode.

2. Target-Organism Pathway Definition

Process included. The reviewed module is the conserved bacterial pathway segment that converts glycolate and glyoxylate into glycerate, a metabolite that re-enters lower glycolysis/gluconeogenesis. In *P. putida* KT2440 the physiologically relevant sequence is:

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ethylene glycol → glycolaldehyde → glycolate → glyoxylate → tartronate semialdehyde → glycerate
                (Ped ADH/ALDH)      (GlcDEF)      (Gcl)      (GlxR)      (GarK)
                                     └─(Hyi auxil.)
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- **Step 2 — glycolate oxidation:** GlcDEF (PP_3745/3746/3747), a three-subunit FAD/Fe-S glycolate dehydrogenase.
- **Step 3 — glyoxylate condensation:** Gcl (PP_4297), glyoxylate carboligase / tartronate-semialdehyde synthase (EC 4.1.1.47).
- **Step 4 — hydroxypyruvate/tartronate-semialdehyde interconversion:** Hyi (PP_4298), hydroxypyruvate isomerase (EC 5.3.1.22) — auxiliary.
- **Step 5 — tartronate-semialdehyde reduction:** GlxR (PP_4299), tartronate-semialdehyde reductase (EC 1.1.1.60).
- Downstream re-entry: glycerate → 2-phosphoglycerate by glycerate kinase GarK (PP_3178).

Neighboring pathways to keep separate. KEGG [ppu00630](#) is a broad "overview-adjacent" map. The following overlapping processes should be curated as **distinct** modules and not merged into the glycolate/glyoxylate-assimilation module:

- **Glyoxylate shunt** (isocitrate lyase AceA / malate synthase GlcB; KEGG ppu00620/ppu00640) — a distinct anaplerotic route that *consumes* glyoxylate but is not part of the glycolate→glycerate branch.
- **Glycine/serine and C1 metabolism** (glyA1/glyA2, glycine cleavage system gcv, purU deformylases; ppu00785/ppu00670/ppu04981).
- **TCA cycle / aconitase / citrate synthase** (acnA, acnB, gltA, prpC; ppu00020).
- **Nitrogen assimilation / glutamine synthetase family** (glnA, PP_3148, PP_4399, PP_4547, puuA, spuB/spuI; ppu00910).
- **Formate/methane metabolism** (fdoGHI, formate dehydrogenase subunits; ppu00680).
- **Thiolase / acetyl-CoA acetyltransferase family** (bktB, yqeF, PP_0582, PP_2215, PP_3355; ppu00900).

Alternate names / database definitions. The reviewed segment is variously described as "glycolate utilization" (glc genes), the "glyoxylate carboligase (tartronate-semialdehyde) pathway," and the "glycerate pathway." KEGG folds these into map 00630 (Glyoxylate and dicarboxylate metabolism); MetaCyc represents the glycerate branch as the *glycolate and glyoxylate degradation* pathway. The step-1 phosphatase (EC 3.1.3.18) belongs biochemically to **photorespiration** in phototrophs, which is a database-driven inclusion, not a KT2440 physiological reality.

3. Expected Step Model and Satisfiability

Module step	Expected enzyme	KT2440 candidate	Status	Confidence / evidence type
1. 2-phosphoglycolate → glycolate	Phosphoglycolate phosphatase (EC 3.1.3.18)	PP_0416, PP_1907	module_needs_revision / not physiological entry	Homology + biochemical context; direct absence of CBB/ RuBisCO
2. glycolate → glyoxylate	GlcDEF glycolate dehydrogenase (EC 1.1.99.14)	PP_3745/3746/3747	covered	Direct KT2440 experiment (overexpression relieves bottleneck)
3. glyoxylate → tartronate semialdehyde	Glyoxylate carboligase Gcl (EC 4.1.1.47)	PP_4297	covered (GclR-gated)	Direct KT2440 experiment (gcl+glxR sufficient)
4. hydroxypyruvate ⇌ tartronate semialdehyde	Hydroxypyruvate isomerase Hyi (EC 5.3.1.22)	PP_4298	covered / auxiliary	Operon co-transcription; not strictly required
5. tartronate semialdehyde → glycerate	Tartronate-semialdehyde reductase GlxR (EC 1.1.1.60)	PP_4299	covered	Direct KT2440 experiment (gcl+glxR sufficient)
(downstream) glycerate → 2-P-glycerate	Glycerate kinase GarK (EC 2.7.1.165)	PP_3178	covered (re-entry)	Homology + pathway logic
(transport) glycolate/ glyoxylate import	GlcA-type permease	— none in candidates	gap	Absence in candidate set; intracellular supply route

Steps encoded by candidate genes: 2, 3, 4, 5, plus glycerate re-entry (GarK). These constitute a satisfiable core.

Steps missing from metadata but likely present under other names: none of the core catalytic steps are missing; the reductase step has *extra* paralogs rather than a gap. Transport is the genuine unfilled slot (see §5).

Steps probably not expected in this organism: the physiological version of step 1 (2-phosphoglycolate salvage as the glycolate source) is not expected in this RuBisCO-lacking heterotroph.

4. Candidate Genes and Evidence

High-confidence core module genes

GlcDEF — PP_3745 (glcD), PP_3746 (glcE), PP_3747 (glcF); EC 1.1.99.14 (Finding F002). This three-subunit FAD-linked / FAD-binding / iron-sulfur glycolate dehydrogenase performs the committed glycolate→glyoxylate oxidation. Franden et al. 2018 showed by direct engineering in KT2440 that overexpression of the *glcDEF* operon **removes the glycolate bottleneck** during ethylene-glycol metabolism and minimizes accumulation of toxic glycolaldehyde/glyoxal intermediates ([PMID: 29885475](#)). Mückschel et al. 2012 independently documented temporal accumulation of glycolic and glyoxylic acid in KT2440, consistent with GlcDEF being the flux-controlling glycolate-oxidizing activity ([PMID: 23023748](#)). **Evidence type: direct KT2440 experiment. Curation: mark step 2 covered.** No paralog ambiguity; these three loci are the unambiguous complex.

Gcl — PP_4297; glyoxylate carboligase / tartronate-semialdehyde synthase; EC 4.1.1.47 (Findings F001, F003). Gcl condenses two glyoxylate molecules to tartronate semialdehyde, committing glyoxylate carbon to the glycerate branch. Franden et al. 2018 demonstrated by directed overexpression that expressing **only *gcl* plus *glxR* was sufficient to confer growth on ethylene glycol**, with full-operon expression improving growth ([PMID: 29885475](#)). **Evidence type: direct KT2440 experiment. Curation: covered, but GclR-gated.**

GlxR — PP_4299; tartronate-semialdehyde reductase; EC 1.1.1.60 (Findings F001, F005). GlxR reduces tartronate semialdehyde to glycerate, the module's exit metabolite. Validated as the functional reductase in the same *gcl+glxR* sufficiency experiment (PMID: 29885475). **Evidence type: direct KT2440 experiment. Curation: covered.** This is the validated reductase; do not let the broad-EC paralogs (below) dilute the assignment.

Hyi — PP_4298; hydroxypyruvate isomerase; EC 5.3.1.22 (Finding F001). Co-transcribed within the *gcl* operon. Provides tartronate-semialdehyde/hydroxypyruvate interconversion, linking hydroxypyruvate metabolism to the GlxR reduction step. It is **auxiliary** — not strictly required for growth (*gcl+glxR* alone suffice) — but part of the physiological operon. **Curation: covered / auxiliary.**

Operon architecture (Finding F001). Franden et al. 2018 established by qRT-PCR that the four genes genomically proximal to *gcl* (*hyi*, *glxR*, *ttuD*, *pykF*) are **co-transcribed as a single operon** with *gcl* (PMID: 29885475). This operon structure is the strongest single piece of KT2440-specific evidence for the module and should anchor its curation.

GarK — PP_3178; glycerate kinase; EC 2.7.1.165 (Finding F005). Downstream re-entry step: phosphorylates glycerate to 2-phosphoglycerate, feeding lower glycolysis. Evidence is homology + pathway logic; not directly tested here but mechanistically required.

Regulatory context (not a catalytic candidate but curation-critical)

GclR — represses the *gcl*/glyoxylate-carboligase branch (Finding F003). Li et al. 2019 identified the transcriptional regulator **GclR** as the repressor of the glyoxylate carboligase pathway; **loss-of-function *gclR* mutations were required** to enable growth on ethylene glycol as sole carbon source in adaptive laboratory evolution (PMID: 31166064). Mückschel et al. 2012 corroborated the regulatory state at the pathway level: in *P. putida* JM37, Gcl, malate synthase (GlcB), and isocitrate lyase (AceA) were all induced by ethylene glycol/glyoxylate, whereas **KT2440 showed induction of AceA only** — the *gcl* branch was transcriptionally silent in wild-type KT2440 (PMID: 23023748). This explains the apparent paradox: the genes are present and functional, but the wild-type pathway is off. **Curation: annotate the module as "covered but conditionally active / GclR-repressed."**

Lower-confidence / over-annotated candidates

Step-1 phosphatases — PP_0416 (phosphoglycolate phosphatase, EC 3.1.3.18), PP_1907 (HAD hydrolase) (Finding F004). PGP dephosphorylates 2-phosphoglycolate to glycolate (PMID: 38861076), but 2-phosphoglycolate is generated primarily by the RuBisCO oxygenase

reaction in CBB-cycle / photorespiratory organisms (PMID: 38593075). KT2440 lacks RuBisCO and the CBB cycle; its only endogenous 2-phosphoglycolate is the minor DNA-repair-derived pool. In the characterized KT2440 route glycolate comes from glycolaldehyde oxidation, not from 2-phosphoglycolate. **Curation: PP_0416 retained as housekeeping/detoxification phosphatase; step 1 flagged not the physiological entry.**

Reductase paralogs — ghrB (PP_1261), hprA (PP_0762), ttuD (PP_4300) (Finding F005). These three carry broad or overlapping EC annotations (ghrB: EC 1.1.1.215/1.1.1.79/1.1.1.81, primary bucket ppu00030; hprA: glycerate dehydrogenase, ppu00680; ttuD: hydroxypyruvate reductase EC 1.1.1.81, ppu00561) that could *nominally* satisfy the tartronate-semialdehyde/hydroxypyruvate reduction step. GlxR is the validated reductase; these paralogs are **likely over-annotated** for this module. Note the specific conflict in **ttuD (PP_4300)**: it is co-transcribed in the *gcl* operon and named for the tartrate-utilization glycerate *kinase*, yet the metadata annotates it as hydroxypyruvate *reductase* — an unresolved name/function conflict warranting individual review.

Clear map artifacts. PP_0094 (5'-nucleotidase, EC 3.1.3.5) has **no role** in glycolate/glyoxylate assimilation and is a KEGG-map artifact. The bulk of the remaining 53 candidates (glyoxylate-shunt, TCA, C1, nitrogen, thiolase enzymes) are neighboring-bucket noise (Finding F006).

5. Gaps, Ambiguities, and Likely Over-Annotations

Candidate-list composition (Finding F006). Of 62 candidates, only 9 are **ppu00630**-primary (PP_0094, PP_0416, PP_1907, glcD/E/F, gcl, hyl, glxR), and only ~7 **form the true module core** (glcDEF, gcl, hyl, glxR + PGP step). The other 53 belong to other primary KEGG buckets and are pulled in because map 00630 is a broad hub overlapping glycine/serine, TCA, the glyoxylate shunt, C1, nitrogen, and thiolase metabolism. Curators should down-weight the list to the core seven and treat the rest as cross-map contamination.

Transport gap (Finding F007). The *E. coli glc* system includes a glycolate permease (GlcA, part of the *glcDEFGBA* cluster). In KT2440 there is **no annotated glycolate/glyoxylate permease among the candidates**; the *glcDEF* operon lacks an adjacent transporter and malate-synthase gene (KT2440 *glcB*, PP_0356, is a separate ppu00620 locus). Because glycolate is generated **intracellularly** from ethylene-glycol-derived glycolaldehyde, a dedicated import step is not strictly required for that carbon-entry mode. Whether exogenous glycolate uptake occurs, and via which transporter, is **unresolved** for KT2440 and is the module's genuine open gap.

Regulatory conditionality. The module is satisfiable but **off in the wild type**. Any "covered" call must carry the GclR-repression caveat; the module is fully active only in *gclR* de-repressed backgrounds.

Step-1 mismatch. The generic module boundary that places a 2-phosphoglycolate phosphatase as the glycolate entry is **wrong for this organism** and reflects a phototroph-centric definition.

6. Module and GO-Curation Recommendations

Module step	Recommended status	Rationale
Step 1 — phosphoglycolate → glycolate	module_needs_revision (not_expected_in_target_taxon as physiological entry)	No RuBisCO/CBB in KT2440; PP_0416 is housekeeping only
Step 2 — glycolate → glyoxylate (GlcDEF)	covered	Direct KT2440 experiment
Step 3 — glyoxylate → tartronate semialdehyde (Gcl)	covered (flag GclR-repressed)	Direct KT2440 experiment
Step 4 — hydroxypyruvate isomerization (Hyi)	covered / auxiliary	Operon member; non- essential for growth
Step 5 — tartronate semialdehyde → glycerate (GlxR)	covered	Direct KT2440 experiment
Transport (glycolate import)	gap (candidate_uncertain)	No GlcA-type permease in candidate set
Reductase paralogs (ghrB/ hprA/ttuD)	candidate_uncertain / likely over- annotated	Broad EC mappings; GlxR is validated reductase

Module boundary revisions. (1) Redefine the module entry for heterotrophic Pseudomonads so that glycolate originates from glycolaldehyde oxidation rather than 2-phosphoglycolate dephosphorylation, or explicitly mark the phosphatase step as lineage-conditional (present

only in CBB/photorespiratory organisms). (2) Add a **regulatory annotation node (GclR)** to capture the conditional-activity state, since satisfiability here is regulatory, not genetic. (3) Add a **transport slot** marked as an open gap.

GO-curation notes. GlcDEF, Gcl, and GlxR have strong experimental support for specific GO molecular-function and biological-process terms (glycolate catabolic process; tartronate-semialdehyde synthase activity; 2-hydroxy-3-oxopropionate reductase activity). Avoid propagating the broad multi-EC reductase GO terms from ghrB to the module reductase role. No new GO term requests appear strictly necessary, but a **GO annotation extension** capturing GclR regulation (or a "conditionally active/repressed" qualifier) would improve fidelity.

7. Genes to Promote to Full fetch-gene Review

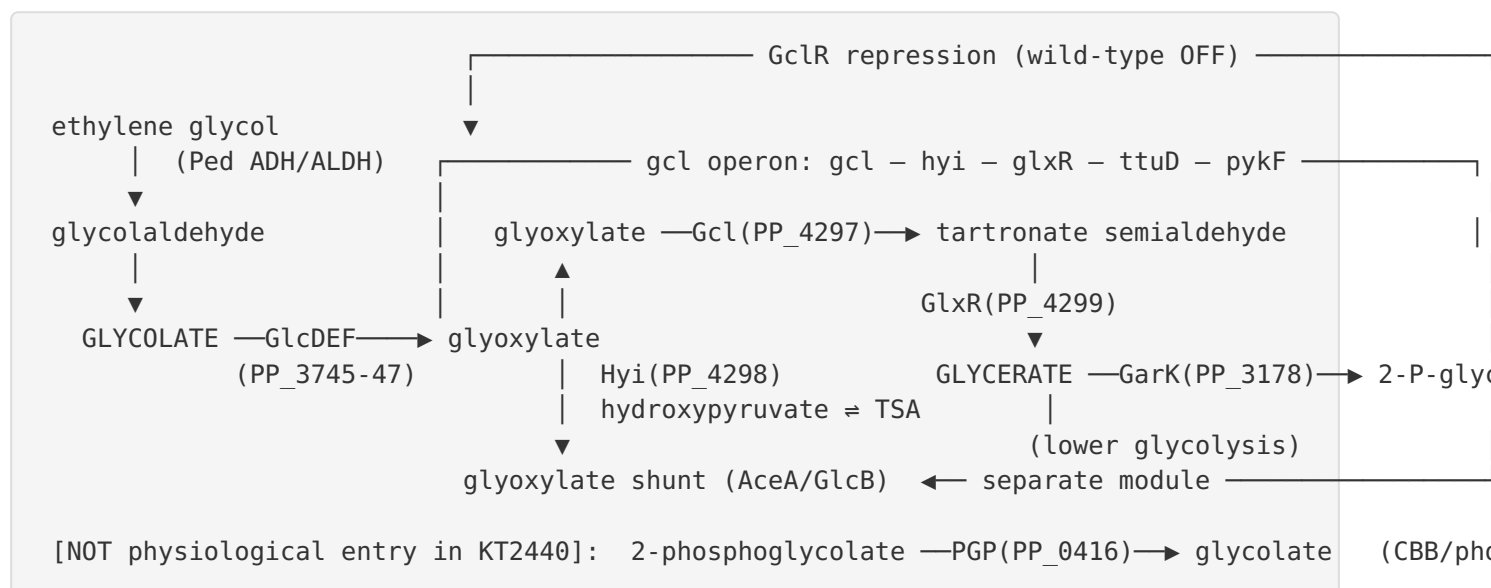
1. **PP_4300 (ttuD)** — resolve the name/function conflict (tartrate-utilization glycerate kinase vs. annotated hydroxypyruvate reductase EC 1.1.1.81); it is a *gcl*-operon member and its true role affects module wiring.
2. **PP_0416 (PGP, EC 3.1.3.18)** — confirm housekeeping/detoxification role and formally decouple it from the physiological glycolate-entry annotation.
3. **PP_1261 (ghrB)** and **PP_0762 (hprA)** — assess whether their broad EC/GO mappings should be trimmed to avoid over-propagation onto the module reductase step.
4. **GlcA-type permease search** — a targeted genome-wide review to determine whether a glycolate/glyoxylate transporter exists outside the candidate set (the module's real gap).
5. **GclR regulator locus** — promote for annotation as the repressor controlling module activity (curation-critical even though non-catalytic).

The validated core (glcDEF, gcl, glxR, hyi) does not require further gene-level review beyond confirming loci — the experimental evidence is direct and sufficient.

8. Evidence Base

Paper	PMID	Contribution	Evidence strength for KT2440
<i>Engineering P. putida KT2440 for efficient ethylene glycol utilization</i> (Franden et al. 2018)	29885475	Operon structure of <i>gcl-hyI-glxR-ttuD-pykF</i> ; <i>gcl+glxR</i> sufficiency for growth; <i>gclDEF</i> overexpression relieves glycolate bottleneck	Direct, strain-specific
<i>Laboratory evolution reveals the metabolic and regulatory basis of ethylene glycol metabolism by P. putida KT2440</i> (Li et al. 2019)	31166064	GclR identified as repressor of the glyoxylate carboligase pathway; <i>gclR</i> loss-of-function required for growth	Direct, strain-specific
<i>Ethylene glycol metabolism by Pseudomonas putida</i> (Mückschel et al. 2012)	23023748	Temporal glycolate/glyoxylate accumulation; KT2440 induces AceA only (<i>gcl</i> branch silent)	Direct, strain-specific
<i>Determination of Phosphoglycolate Phosphatase Activity...</i>	38861076	PGP dephosphorylates 2-phosphoglycolate to glycolate (photorespiratory context)	Biochemical context (not KT2440)
<i>Removal of phosphoglycolate in hyperthermophilic archaea</i>	38593075	2-Phosphoglycolate salvage linked to CBB cycle / oxygenic photosynthesis	Contextual (supports step-1 exclusion)
<i>Ethylene glycol metabolism in Ideonella sakaiensis</i>	36289066	<i>I. sakaiensis</i> EG enzymes are homologs of KT2440 Ped enzymes — confirms upstream glycolaldehyde route	Related organism, supports upstream supply

How the evidence coheres. The three KT2440-specific papers converge on a single narrative: the module is genetically complete and biochemically functional (Franden 2018), the wild-type pathway is regulated off (Mückschel 2012; Li 2019), and de-repression via *gclR* unlocks growth



- **No direct enzymology performed here** — conclusions rest on published KT2440 genetics/physiology plus annotation reasoning, not new wet-lab data.
- **Transporter identity unknown** — whether KT2440 imports exogenous glycolate/glyoxylate, and via which permease, is unresolved.
- **ttuD name/function conflict unresolved** — kinase vs. reductase annotation needs experimental disambiguation.

- **Reductase paralog specificities** (ghrB, hprA) toward tartronate semialdehyde have not been kinetically compared with GlxR in KT2440.
- **Quantitative regulatory thresholds** — the exact GclR effector/derepression conditions in the wild type are only partly defined.

11. Proposed Follow-up Experiments / Actions

1. **Genome-wide permease search** (targeted `fetch-gene`) for a GlcA/glyoxylate-transporter candidate outside the current bucket; test exogenous glycolate/glyoxylate uptake in KT2440.
 2. **Disambiguate PP_4300 (ttuD)** by enzymatic assay (glycerate kinase vs. hydroxypyruvate reductase) and by phenotyping a clean deletion in a *gclR*-derepressed background.
 3. **Reductase paralog comparison** — kinetically test GlxR vs. ghrB/hprA on tartronate semialdehyde to confirm GlxR is the physiological module reductase and trim over-propagated EC/GO terms.
 4. **Confirm PP_0416 housekeeping role** — assess phenotype on glycolate/EG to verify it is dispensable for assimilation (detoxification-only).
 5. **Curator actions:** mark steps 2/3/5 covered (GclR caveat), step 4 auxiliary-covered, step 1 module_needs_revision, transport gap; add a GclR regulatory node; down-weight the 53 non-core candidates as cross-map noise.
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Prepared as a species-aware curation review. Direct KT2440 evidence is available for the core oxidation and glyoxylate-to-glycerate steps; step-1 exclusion and transport/paralog calls are inferences from pathway logic, comparative genomics, and biochemical context and are flagged accordingly.
